ARTHROPLASTY · NEET-SS / FET / INI-SS

TKA balancing and alignment

1. Alignment concepts

- **Mechanical axis of the lower limb:** centre of the femoral head to the centre of the ankle; normally passes **just medial to the centre of the knee** (about 0-3 degrees varus overall).
- **Anatomical vs mechanical femoral axis:** differ by about **5-7 degrees (valgus)**; the **distal femoral cut is made at about 5-7 degrees valgus to the anatomical axis** (perpendicular to the mechanical axis).
- **Joint line:** the native tibial plateau is in **about 3 degrees varus** relative to the tibial mechanical axis; the distal femur in about 3 degrees valgus relative to the femoral mechanical axis; the **joint line is roughly parallel to the floor in stance** (with feet together).
- **Alignment philosophies:**

Philosophy	Principle	Notes
Mechanical alignment (MA)	Femoral and tibial cuts **perpendicular to their mechanical axes** (neutral HKA, joint line horizontal); balance by **soft-tissue releases**	Traditional gold standard; **neutral axis ±3 degrees** associated with better survival (debated with modern implants)
Anatomical alignment (Hungerford)	Joint line at 3 degrees varus, neutral limb	Historical

Philosophy	Principle	Notes
Kinematic alignment (KA)	**Resurface the pre-arthritic joint** (restore native joint lines and HKA), **minimal releases**	Implants positioned within the patient's native alignment; outlier limb alignments; satisfaction similar or slightly better in some trials
Restricted KA / inverse kinematic / functional alignment	KA within safe limits; robotic gap-based adjustments	**CPAK classification** (Coronal Plane Alignment of the Knee: arithmetic HKA and joint line obliquity; 9 phenotypes) describes constitutional alignment

- **Robotic/navigation** TKA reduces alignment outliers; clinical superiority not yet proven.

2. Bone cuts and component rotation

- **Tibial cut:** perpendicular to the tibial mechanical axis in the coronal plane (MA), **posterior slope about 3-7 degrees** (CR knees more; PS knees less, as excess slope can cause instability in flexion/post impingement). Resect about **8-10 mm from the unaffected (higher) side** (measured resection).
- **Distal femoral cut:** about 5-7 degrees valgus to the anatomical axis (intramedullary guide entry point just medial and anterior to the PCL origin); resection equal to the implant thickness (about 9-10 mm).
- **Femoral component rotation (critical for patellar tracking and flexion gap balance):**
  - **Transepicondylar axis (TEA)** (lateral epicondyle to the **medial epicondylar sulcus**: surgical TEA): **most reliable** rotational landmark.

  - **Whiteside line (AP axis):** from the deepest part of the trochlear groove to the centre of the intercondylar notch; femoral component **perpendicular** to it (unreliable in trochlear dysplasia).
  - **Posterior condylar axis (PCA):** femoral component **externally rotated about 3 degrees** relative to it (because the medial posterior condyle is larger); **beware a hypoplastic lateral condyle in valgus knees** (would internally rotate the component: use TEA/Whiteside).
  - **Gap balancing technique:** rotate the femoral component so the **flexion gap is rectangular** after the tibial cut with balanced ligaments.
- **Tibial component rotation:** aligned to the **medial third of the tibial tubercle** (Akagi line: PCL attachment to the medial border of the patellar tendon at its tibial attachment).
- **Internal rotation of femoral or tibial components** (combined) causes **patellar maltracking, lateral subluxation, anterior knee pain, patellar complications and stiffness** (Berger: combined internal rotation 1-4 degrees = tilt; 3-8 = subluxation; 7-17 = dislocation/failure). Assess with **CT**.
- **Anterior femoral notching** (anterior cortex): increases **supracondylar periprosthetic fracture** risk (about 30-40% reduction in torsional strength).
- **Femoral sizing: anterior referencing** (cuts the posterior condyles by the measured amount; downsizing changes the flexion gap: may loosen flexion) vs **posterior referencing** (restores posterior condylar offset; risk of anterior notching or overstuffing the patellofemoral joint).

3. Gap balancing

- **Extension gap** (distal femoral cut + tibial cut) and **flexion gap** (posterior femoral cut + tibial cut) must be **equal and rectangular**.

- **The tibial cut affects both gaps; the distal femoral cut affects only the extension gap; the posterior femoral cuts/femoral size affect only the flexion gap.**

Problem	Flexion gap	Extension gap	Solution
Tight in both	Tight	Tight	**Cut more tibia**
Loose in both	Loose	Loose	**Thicker polyethylene insert** (or augments)
Tight in extension, OK in flexion	Balanced	Tight	**Release posterior capsule / remove posterior osteophytes**, then **resect more distal femur** (about 2 mm)
Loose in extension, OK in flexion	Balanced	Loose	**Distal femoral augmentation** (add distal femur), thicker insert + smaller/posterior cut (combined)
Tight in flexion, OK in extension	Tight	Balanced	**Downsize the femoral component** (anterior referencing) or **increase tibial slope**; release PCL (CR)
Loose in flexion, OK in extension	Loose	Balanced	**Upsize femur** (posterior augments) or **thicker insert + more distal femoral resection**

- **Joint line:** keep within about **8 mm** of native: **elevation of the joint line** (excess distal femoral resection + thick insert) causes **mid-flexion instability, patella baja (pseudo), patellofemoral problems**.

4. Coronal deformity correction (soft-tissue releases)

Varus knee (commonest)

- **Tight medial structures:** remove **medial osteophytes first** (tibial and femoral), then release **deep MCL (meniscotibial)**, **posteromedial capsule/semimembranosus** (tight in **extension**), **superficial MCL** (subperiosteal from tibia or **pie-crusting**; the **anterior sMCL is tight in flexion**, posterior in extension), **pes anserinus** (rarely). **PCL** release (in CR knees with tight flexion).
- Also: **medial tibial reduction osteotomy** (downsize/lateralise the tibial tray and remove medial bone).
- **Fixed flexion deformity:** remove posterior osteophytes, **posterior capsular release**, gastrocnemius recession; as a last resort extra distal femoral resection.

Valgus knee

- **Tight lateral structures: iliotibial band** (tight in **extension**), **popliteus** (tight in **flexion**), **lateral collateral ligament** (tight in **flexion and extension**), **posterolateral capsule** (extension), lateral head of gastrocnemius, biceps femoris.
- **Order (Ranawat inside-out pie-crusting, Whiteside):** posterolateral capsule, ITB, then popliteus/LCL as needed.
- **Hypoplastic lateral femoral condyle**: do not use the PCA alone for rotation.
- **Peroneal nerve palsy** risk when correcting **severe valgus with flexion contracture** (stretch): **flex the knee post-operatively**, loosen dressings; explore if no recovery.
- **Ranawat valgus types:** I (correctable, medial soft tissue intact), II (fixed, medial attenuated: may need constraint), III (severe osseous deformity with incompetent medial structures: constrained implant).

5. Constraint and design choices

Design	Mechanism	Indications
Cruciate-retaining (CR)	PCL retained; flatter or ultra-congruent inserts	Intact, functional PCL; less bone resection; **paradoxical anterior femoral sliding**
Posterior-stabilised (PS)	**Cam-post** substitutes the PCL (**femoral rollback**)	**Absent/incompetent PCL**, **patellectomy**, inflammatory arthritis, large deformity, previous HTO (debated); risks: **post wear/fracture, patellar clunk syndrome** (fibrous nodule at the superior pole catching the box), cam jump (flexion instability)
Medial pivot / ultra-congruent	Highly conforming medial compartment	Mimics native kinematics
Varus-valgus constrained (CCK, non-linked)	Tall, wide post in a deep box	**MCL/LCL incompetence**, severe valgus, flexion-extension mismatch, revisions
Rotating-hinge (linked)	Hinged femur-tibia linkage with rotation	**Global ligamentous deficiency**, severe bone loss, distal femoral replacement (tumour/fracture), hyperextension instability, **neuromuscular recurvatum** (polio)

- **Use the least constraint needed** (more constraint = more stress transfer to fixation interfaces = loosening).
- **Mobile-bearing vs fixed-bearing**: no proven difference in survival; spinout risk with mobile bearings.

- **Patellar resurfacing:** reduces **reoperation for anterior knee pain** (meta-analyses, KAT trial: little functional difference); inflammatory arthritis favours resurfacing. **Patellar thickness should be restored** (avoid overstuffing). **Patellar complications:** maltracking (component internal rotation, lateral placement of femoral/tibial components, medial placement of patellar button is preferred), **fracture** (over-resection, avascular after lateral release), **avascular necrosis**, **patellar clunk**, loosening, tendon rupture.
- **Patellar tracking: "no thumb" test**; lateral retinacular release if needed (protect the **superior lateral genicular artery**).

6. Complications of TKA (non-infective)

- **Stiffness/arthrofibrosis** (flexion under about 90 degrees at 6-12 weeks): **manipulation under anaesthesia (within about 6-12 weeks)**, arthrolysis, revision.
- **Instability: flexion instability** (flexion gap larger than extension; recurrent effusions, PS cam jump), **mid-flexion instability**, **genu recurvatum**, global instability: revise with balanced gaps/increased constraint.
- **Aseptic loosening/osteolysis**, **polyethylene wear**.
- **Periprosthetic fractures: supracondylar femur (Lewis-Rorabeck I undisplaced, II displaced + stable prosthesis (ORIF: locking plate or retrograde nail if open box), III loose prosthesis (revision/DFR))**; **tibial (Felix)**; **patella (Goldberg/Ortiguera)**.
- **Peroneal nerve palsy** (valgus, flexion contracture, epidural anaesthesia), **vascular injury** (popliteal artery behind the posterior capsule: **posterolateral corner, about 9 mm from the posterior tibia in flexion**).
- **Wound complications: use the most lateral previous incision** (lateral skin flap less perfused: blood supply from the medial side), skin bridges at least about 5-6 cm.

- **Extensor mechanism disruption:** patellar tendon rupture (**worst; augment with allograft or synthetic mesh, medial gastrocnemius flap**), quadriceps rupture.
- **VTE**, **persistent pain** (about 10-20% dissatisfied after TKA: highest predictors are pre-operative expectations, pain catastrophising, mild radiographic OA).
- **Unexplained painful TKA** work-up: **exclude infection first (ESR/CRP, aspiration)**, then instability, malrotation (CT), loosening, patellar issues, referred pain (hip, spine), CRPS, neuroma.

> Exam pearl: **Tibial cut affects both gaps; distal femoral cut affects extension only; posterior femoral cut/femoral size affects flexion only. Tight both = cut more tibia; loose both = thicker insert.**

> Exam pearl: **Femoral rotation: TEA most reliable; 3 degrees external to the posterior condylar axis; Whiteside line perpendicular. Internal rotation of components = patellar maltracking. Tibial rotation: medial third of the tubercle.**

> Exam pearl: **Varus: medial osteophytes, deep MCL, posteromedial capsule/semimembranosus, then superficial MCL. Valgus: ITB (extension), popliteus (flexion), LCL (both); peroneal palsy risk with valgus + flexion contracture.** Least constraint necessary; rotating hinge for global ligament loss.

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.